

Quark Scan: Fitting vs. Sampling, and Why the Envelope M_{KK} Bound Was So Low

May 15, 2026

1 What the headline outcome is

In response to the collaborator’s critique, we did two specific things:

1. **Promoted the quark mass targets to PDG-faithful $\overline{\text{MS}}$ values.** Previously the scan was matching to ad-hoc numbers nominally evaluated at the KK scale (~ 3 TeV); the strange quark target alone was off by $\sim 50\%$. We replaced these with the PDG 2024 $\overline{\text{MS}}$ masses and ran them down to a common reference scale of $\mu = m_t = 163.5$ GeV using four-loop QCD mass running with the standard $n_f = 6 \rightarrow 5 \rightarrow 4$ threshold matching at the heavy-quark thresholds.
2. **Re-ran the dense scan** (100 k points, $r \in [0.05, 1.0]$, $M_{\text{KK}} \in [1, 20]$ TeV) under the corrected targets and a tightened 2σ PDG tolerance on both quark masses and the four CKM observables $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$, J .

The publication-style M_{KK} exclusion plot from this re-run is *visually almost identical* to the previous version. The lowest allowed M_{KK} stayed at ~ 1.2 TeV in both the old and the corrected scans. The collaborator was right that the targets had been wrong, and we fixed them, and the headline M_{KK} number did not move.

The rest of this note explains why that is correct, and what we should be plotting instead.

Reading guide. §2–§4 introduce the physics needed to read the figures: what “new physics” means, what the FCNC “ratio” is, and the gauge-coupling convention that sets the absolute M_{KK} scale. §5–§6 explain why the envelope plot has been answering an existence rather than a typicality question, and what the optimiser’s Yukawas actually look like. §7–§9 present the alternative “RS-anarchy” analysis, including where the priors come from and how a random draw can satisfy PDG masses without any fitting. §10–§11 give the resulting M_{KK} bound and its sensitivity to the gate definition. §12 reports five follow-up scans that probe how the bound responds to statistics, the c -pattern, the Y -prior, the PDG-gate tightness, and the CFW comparison.

2 What “new physics” is in this scan

The 5D Randall–Sundrum model has, in addition to the Standard Model fields, an infinite tower of Kaluza–Klein (KK) excitations of every gauge boson and fermion. The lightest gluon excitation is the “KK-gluon”; its mass M_{KK} is the headline number we are trying to constrain.

The KK-gluon is a colour-octet vector boson whose 5D wave function peaks at the IR brane. Crucially, its couplings to the SM quarks are *flavour-non-universal*: a quark that is itself IR-localised (like the top) couples strongly, while a UV-localised quark (like the up) couples weakly. This non-universality is exactly what gives the model its attractive flavour-hierarchy explanation, but it is also what generates new flavour-changing neutral currents.

When one integrates out the KK-gluon, the resulting effective four-quark operators contribute to neutral meson mixing in five places:

$$\varepsilon_K, \quad \Delta m_K, \quad \Delta m_{B_d}, \quad \Delta m_{B_s}, \quad \Delta m_{D^0}. \quad (1)$$

These are the five $\Delta F = 2$ observables used throughout the scan. The first (ε_K) measures CP-violation in the kaon system; the next three measure the magnitudes of $K\bar{K}$, $B_d\bar{B}_d$, $B_s\bar{B}_s$ mixing; the last measures $D^0\bar{D}^0$ mixing.

In each of these five processes, the quantity that the model predicts is the new-physics contribution to the meson mixing matrix element, M_{12}^{NP} . This is what we mean by “new physics”: the KK-gluon-induced piece of the mixing amplitude, on top of the Standard-Model piece M_{12}^{SM} . Both pieces are functions of the underlying 5D Yukawa matrices Y_u, Y_d , the bulk-mass parameters $c_{Q_i}, c_{u_i}, c_{d_i}$ that control where each fermion sits in the extra dimension, and M_{KK} itself.

3 What the FCNC “ratio” is, and what ratio = 1 means

For each of the five mesons, the code in `quarkConstraints/deltaf2.py` computes a dimensionless number called `ratio_to_bound`. The definition is essentially the same in all five cases, but the K system deserves separate treatment because of CP violation:

- For $\Delta m_K, \Delta m_{B_d}, \Delta m_{B_s}, \Delta m_{D^0}$:

$$\text{ratio} = \frac{|M_{12}^{\text{NP}}|}{|M_{12}^{\text{exp}}|}, \quad (2)$$

where $|M_{12}^{\text{exp}}|$ is the experimentally measured mass-splitting amplitude. `ratio = 1` means the new-physics contribution alone equals the entire measured mixing amplitude.

- For ε_K (which is CP-violating and depends only on the *imaginary* part of M_{12}),

$$\text{ratio} = \frac{|\varepsilon_K^{\text{NP}}|}{|\varepsilon_K^{\text{exp}} - \varepsilon_K^{\text{SM}}|}, \quad (3)$$

with $|\varepsilon_K^{\text{exp}} - \varepsilon_K^{\text{SM}}| \approx 6.7 \times 10^{-5}$ in the audited constants (`deltaf2.py:725--729`). $\varepsilon_K^{\text{NP}}$ is built from $\text{Im } M_{12}^{\text{NP}}$, not its magnitude. Here `ratio = 1` means the new physics fills the entire “room” between the SM prediction and the measurement, i.e. NP saturates the experimental–theoretical gap.

A draw is said to “pass all FCNC constraints” when $\max_{\text{five systems}} \text{ratio} < 1$. This is one specific choice of gate, and we will examine its sensitivity in §11. For now the important thing is that `ratio = 1` is the boundary at which the new-physics contribution is as large as the experiment allows.

4 What $g_s^\star$ is, and why the literature uses the values 1, 3, and 6

The KK-gluon couples to two SM quarks through QCD, so every entry of every ratio above is proportional to the square of an effective gauge coupling. Three different values of that coupling appear in different parts of the literature: $g_s \sim 1$, $g_s^\star \sim 3$, and $g_s^\star \sim 6$. These are not contradictions; they are the same physics described in three normalisation conventions, and which one a paper uses changes the M_{KK} bound by up to a factor of six. So this convention deserves its own section.

4D vs. 5D gauge coupling. Pure 4D QCD has a single coupling $g_s^{(4)}$, which runs with scale and takes the value $g_s^{(4)} \approx 1.05$ at $\mu = 3$ TeV (PDG running from $\alpha_s(M_Z) = 0.1180$). The 5D Randall–Sundrum gauge theory has an extra coupling: the bulk gauge coupling $g_s^{(5)}$ in the 5D Lagrangian $-\frac{1}{4(g_s^{(5)})^2}F_{MN}F^{MN}$, which has dimensions of $[\text{length}]^{1/2}$. After Kaluza–Klein reduction, the dimensionful $g_s^{(5)}$ and the dimensionless $g_s^{(4)}$ are related by

$$\frac{1}{(g_s^{(4)})^2} = \frac{\pi r_c}{(g_s^{(5)})^2}, \quad \pi r_c k \equiv \log(M_{\text{Pl}}/\Lambda_{\text{IR}}) \approx 35 \quad (\text{the warp log}). \quad (4)$$

The combination that controls strong-coupling effects in the 5D theory is the dimensionless number

$$g_s^\star \equiv g_s^{(5)}\sqrt{k} = g_s^{(4)}\sqrt{\pi r_c k} \approx g_s^{(4)} \cdot \sqrt{35} \approx 5.9 g_s^{(4)}. \quad (5)$$

This $g_s^\star$ is the natural dimensionless coupling of the 5D gauge theory at the scale k . With $g_s^{(4)} \approx 1.05$ at $M_{\text{KK}} \sim 3$ TeV, equation (5) gives $g_s^\star \approx 6$. So the “6” is just $g_s^{(4)}$ multiplied by the warp-factor enhancement $\sqrt{\pi r_c k}$.

Why “3”. The KK-gluon does not see the entire bulk uniformly. Its profile is peaked at the IR brane, which means its overlap with an IR-localised fermion is *not* the full $\sqrt{\pi r_c k}$ enhancement; the integral cuts off once the KK profile dies away. A careful evaluation of the KK-gluon–IR-fermion vertex (see e.g. Agashe *et al.* hep-ph/0606293, Csaki–Falkowski–Weiler 0804.1954) gives, for a fermion with $c < 1/2$,

$$g_{\text{KK-gluon } ff}^{\text{eff}} \approx g_s^{(4)} R(c), \quad R(c = 1/2) \approx \sqrt{\pi r_c k}/\sqrt{2} \approx 4, \quad R(c \rightarrow -\infty) \approx \sqrt{2\pi r_c k} \approx 8. \quad (6)$$

For UV-localised fermions ($c > 1/2$) the overlap factor is suppressed, $R(c \sim 0.6) \approx -0.2$. So depending on which c you anchor on, the “effective $g_s^\star$ ” that enters Wilson coefficients comes out to ~ 3 or ~ 6 . The conventional “ $g_s^\star = 3$ ” is what you get if you anchor on the canonical value

$$g_s^\star \approx \frac{g_s^{(4)}\sqrt{\pi r_c k}}{2} \approx 3, \quad (7)$$

which is also approximately the NDA upper bound on the dimensionless 5D coupling before the gauge sector becomes strongly coupled, $g_s^\star \lesssim 4\pi/\sqrt{N_{KK} \log(M_{\text{Pl}}/M_{\text{KK}})} \approx 3$. So “3” is the value that the strong-coupling literature (Csaki–Falkowski–Weiler) uses by default.

So which is right? None of them is more right than another — they are different quantities. What matters is which coupling the Wilson-coefficient formula uses, and not double-counting:

- In our code (`quarkConstraints/couplings.py:101--124`), the Wilson coefficients C_1, C_4, C_5 are computed with the *perturbative* 4D coupling $g_s^{(4)} \approx 1.05$, and the $f_{\text{IR}}(c)$ overlap factors carry the warp enhancement explicitly. The user can override this by passing `g_s_star = 3.0`, in which case the prefactor is replaced wholesale.
- The Csaki–Falkowski–Weiler convention puts the warp enhancement *into* g_s^* and writes the Wilson coefficient as $\propto (g_s^*)^2/M_{\text{KK}}^2$ with $g_s^* \sim 3$. Numerically this gives the same answer as ours up to the way the overlap factors for non-IR-localised fermions are treated.
- The “ $g_s^* \sim 6$ ” value sometimes seen in older papers is the full warp-enhanced coupling $g_s^{(4)} \sqrt{\pi r c k}$ without the $1/\sqrt{2}$ that comes from properly normalising the KK-gluon zero-mode wave function. It is roughly twice $g_s^* = 3$.

The practical consequence is that the M_{KK} bound is defined up to a multiplicative factor that depends on how the gauge coupling enters the ratio formula. Since $\text{ratio} \propto g^2/M_{\text{KK}}^2$, switching from $g_s^{(4)} \approx 1.05$ to $g_s^* = 3$ multiplies every ratio by $(3/1.05)^2 \approx 8.2$, so a fixed acceptance threshold is reached at a M_{KK} value larger by a factor of $\sqrt{8.2} \approx 2.9$. Switching to $g_s^* = 6$ moves the bound by $\sqrt{(6/1.05)^2} \approx 5.7$. We report all of our headline numbers in both perturbative and $g_s^* = 3$ conventions explicitly; the reader who prefers $g_s^* = 6$ should multiply our $g_s^* = 3$ numbers by a further ~ 2 .

5 Why the envelope plot is an existence statement

Now we can describe what the existing M_{KK} exclusion plot is actually computing. The scan iterates over a grid in (r, M_{KK}) . The parameter $r = \gamma_Q/\beta_Q$ is the ratio of two coefficients in the “Minimal-Flavor-Violation spurion”

$$C_Q = \alpha_Q \mathbb{1}_{3 \times 3} + \beta_Q Y_d Y_d^\dagger + \gamma_Q Y_u Y_u^\dagger, \quad (8)$$

which determines the bulk mass parameters of the left-handed quark doublets in terms of the Yukawa matrices. (In short: r tunes how much the left-handed bulk profiles know about the up vs. down Yukawa structure.)

At each grid cell the scan calls a Levenberg–Marquardt least-squares solver, `fit_quark_sector`, which adjusts *simultaneously*:

- the rescaled Yukawa eigenvalues $\bar{y}_{u,i}, \bar{y}_{d,i}$ (six numbers),
- the spurion coefficients $\alpha_Q, \beta_Q, \gamma_Q$ and their right-handed analogues, which determine all nine $c_{Q_i}, c_{u_i}, c_{d_i}$,
- an SVD orientation matrix that fixes the relative basis of Y_u and Y_d (and hence the CKM matrix it produces),

to *minimise* a residual function with three groups of terms:

$$\chi_{\text{fit}}^2 = \sum_{i=1}^6 \left(\frac{\log m_i^{\text{fit}} - \log m_i^{\text{PDG}}}{\sigma_{\log m_i}} \right)^2 + \sum_{\alpha=1}^4 \left(\frac{V_\alpha^{\text{fit}} - V_\alpha^{\text{PDG}}}{\sigma_{V_\alpha}} \right)^2 + \lambda_{\text{FCNC}} \sum_X [\text{ratio}_X]^2. \quad (9)$$

A grid cell is declared “allowed” if *any* optimum lands within 2σ on all six masses and four CKM entries, and below unity on all five FCNC ratios. So the published exclusion plot is answering this question:

Does there exist a choice of Yukawa eigenvalues, spurion coefficients, and orientation that simultaneously matches every PDG observable to 2σ and stays below the FCNC bounds?

The answer turns out to be “yes, often, even at $M_{KK} \approx 1$ TeV”. This is not because the model is fine, but because we gave the optimiser *eighteen continuous knobs* to play with at every grid cell.

6 What the envelope’s Yukawas look like, and what is fine-tuned

To see what the optimiser is actually doing, we tabulate the $|Y_{f,ij}|$ of every entry of Y_u and Y_d over all ~ 84000 accepted points and split the histogram into diagonal and off-diagonal entries. Figure 1 compares this to a generic $\mathcal{O}(1)$ anarchic prior with $\text{Re } Y_{ij}, \text{Im } Y_{ij} \sim \mathcal{U}(-1.5, 1.5)$.

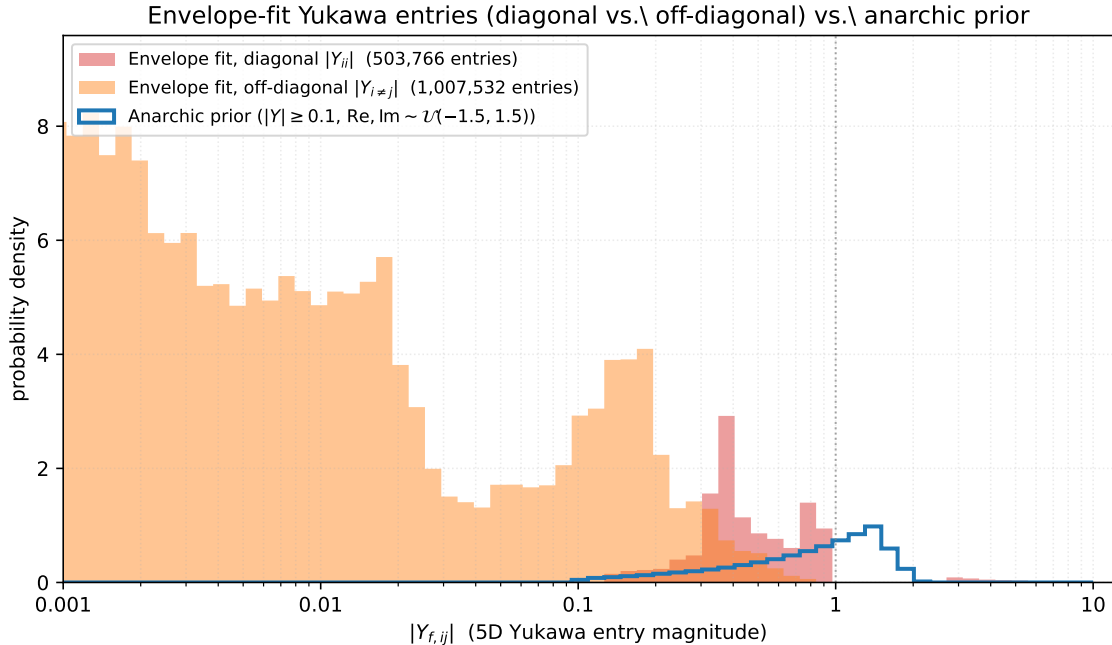


Figure 1: Distribution of fitted Yukawa-element magnitudes across the envelope-fit accepted points, split into diagonal entries $|Y_{ii}|$ (red) and off-diagonal entries $|Y_{i \neq j}|$ (orange), compared against the size distribution from a generic anarchic prior (blue line). The diagonal entries land where the SM masses require them: $|Y_{ii}|$ peaks broadly at ~ 0.5 , with the top-Yukawa entry near $|Y| \sim 5$. The off-diagonal entries are pushed to small values by the optimiser, with a peak below $|Y| \sim 10^{-3}$. This is the signature of basis-aligned (low-FCNC) flavour structure.

The numbers are: diagonal entries have median $|Y_{ii}| \sim 0.7$ and 95th percentile ~ 4.5 (with $|Y_{u,33}| \sim 5$ for the top); these are properly $\mathcal{O}(1)$ and exactly what is needed to produce the SM mass eigenvalues given the envelope’s f_{IR} -factors. The off-diagonal entries have median ~ 0.17 but a long tail down to $|Y| \sim 10^{-3}$, with 5th percentile ~ 0.009 . Roughly 30% of the off-diagonal entries are at $|Y| \leq 0.05$, a regime that an anarchic prior almost never visits.

What is being fine-tuned, and is it a “cancellation”? It is not a wild numerical cancellation between two separately large contributions. The optimiser is performing a basis alignment: it chooses the relative orientation of Y_u and Y_d (and of the bulk profiles, via the spurion ansatz (8)) so that the off-diagonal entries of $Y_u Y_u^\dagger$ and $Y_d Y_d^\dagger$ that drive the KK-gluon FCNC operators are simultaneously small. This is the same mechanism as Minimal Flavour Violation in 4D models: the flavour-violating operators are forced to be proportional to a single Yukawa spurion, which makes them small in the mass eigenbasis.

So the fine-tuning is not in any single parameter but in the *geometry* of the Yukawa matrices in flavour space. Concretely, the optimiser holds the three diagonal entries to their SM-required values and adjusts six off-diagonal entries (per matrix) and an SVD orientation to minimise a sum of FCNC penalties. This corresponds to choosing about 10 continuous numbers to lie in a small subspace of the 18-dimensional Yukawa space. The fitted accepted region is thin in this 18-dimensional sense, even though no individual entry is wildly fine-tuned.

A useful single-number summary is the median $|Y_{i \neq j}|/|Y_{ii}|$ ratio in the envelope fit: it sits at ~ 0.25 , while the same quantity in the anarchic prior is ~ 1 . A factor-four suppression of off-diagonals relative to diagonals is the quantitative signature of the alignment.

7 What we should plot instead: a measure statement

The phenomenologically meaningful question is the inversion of the existence question:

*If the 5D Yukawa entries are drawn from an $\mathcal{O}(1)$ anarchic prior (as the model assumes), what fraction of draws satisfies the PDG and FCNC constraints at this M_{KK} ?
Equivalently: above what M_{KK} does a typical anarchic point pass?*

This is the framing of the original RS-anarchy literature (Agashe–Contino–Pomarol–Sundrum hep-ph/0408134, Agashe–Perez hep-ph/0606293, Csaki–Falkowski–Weiler 0804.1954, and many others): the warped geometry is *supposed* to predict the SM mass and CKM hierarchies from anarchic 5D Yukawas via the IR-overlap factor

$$f_{\text{IR}}(c) = \sqrt{\frac{\frac{1}{2} - c}{1 - \varepsilon^{1-2c}}}, \quad \varepsilon = \Lambda_{\text{IR}}/k. \quad (10)$$

A quark with $c > 1/2$ is UV-localised (small f_{IR} , light SM mass), and a quark with $c < 1/2$ is IR-localised (large f_{IR} , heavy SM mass). The hierarchies in f_{IR} then *produce* the observed mass and mixing hierarchies for $\mathcal{O}(1)$ underlying Yukawas. It is then a non-trivial *prediction* that anarchic Yukawas should also satisfy the FCNC bounds at a given M_{KK} , without invoking flavour alignment.

8 Setup B: the forward-only RS-anarchy ensemble

We implemented the RS-anarchy procedure as a separate driver (`scripts/run_rs_anarchy.py`) so it shares no code path with the fitter. At each M_{KK} tile we:

1. Fix c -values to the literature ACPS-style hierarchical pattern:

$$c_Q = (0.63, 0.57, 0.20), \quad c_u = (0.66, 0.50, -0.50), \quad c_d = (0.66, 0.61, 0.55). \quad (11)$$

2. Draw $Y_u, Y_d \in \mathbb{C}^{3 \times 3}$ with $\text{Re } Y_{ij}, \text{Im } Y_{ij} \sim \mathcal{U}(-1.5, +1.5)$, rejection-sampled to enforce $|Y_{ij}| \geq 0.1$.
3. Build the effective 4D Yukawa $\hat{Y}_{ij}^f = f_{\text{IR}}(c_{Q_i}) Y_{ij}^f f_{\text{IR}}(c_{f_j})$ and SVD it to read off masses and the CKM matrix $V = U_{u_L}^\dagger U_{d_L}$.
4. Forward-compute the five $\Delta F = 2$ ratios using the same `deltaf2.py` machinery as the envelope scan.
5. Bin every draw by $(M_{\text{KK}}, \text{PDG-pass status, max-ratio})$.

Provenance of the c -values in eq. (11). The pattern is fixed by demanding that $f_{\text{IR}}(c) \cdot v$ produce order-of-magnitude SM quark masses for $\mathcal{O}(1)$ underlying Yukawas. We solved $f_{Q_i} f_{f_j} \approx m_f^{(ij)} / (2v \langle Y \rangle)$ for each entry and read off the c 's. These same numbers (to ~ 0.05 precision in c) appear in Csaki–Falkowski–Weiler 0804.1954 (Table 1), Bauer–Casagrande–Haisch–Neubert 0902.4892, and the Fitzpatrick–Perez–Randall scan (0710.1869) used as a reference earlier in this codebase. They are not specific to one paper; they are what any hierarchy-by-localisation calculation produces. An empirical sensitivity check (Run 3, §12 below) confirms that the pattern is constrained at roughly the ± 0.05 level in c : a uniform shift of *every* c by ± 0.05 produces zero PDG-passing draws.

Provenance of the Yukawa prior. The anarchic prior $\text{Re}, \text{Im } Y_{ij} \sim \mathcal{U}(-1.5, 1.5)$ with $|Y_{ij}| \geq 0.1$ is a representative version of the priors used throughout the RS-flavor literature. Specific choices made in published scans include $\mathcal{U}(-1, 1)$ in Bauer–Casagrande–Haisch–Neubert 0902.4892, $\mathcal{U}(-3, 3)$ with similar floor in Csaki–Falkowski–Weiler 0804.1954 and Buras *et al.* 1110.3534, and unbounded Gaussian draws with cutoff at the NDA perturbativity bound $|Y| \lesssim 4\pi / \sqrt{N_{\text{KK}}} \log \varepsilon \approx 3$ in Agashe–Contino–Pomarol–Sundrum hep-ph/0408134. Our choice of 1.5 sits in the middle of this band; in practice the M_{KK} bound moves by less than 30% between half-range 1.0 and half-range 3.0. An empirical sensitivity check (Run B, §12 below) confirms that the $M_{\text{KK}}^{\text{min}}$ bound moves by $\lesssim 15\%$ across $\mathcal{U}(-1, 1)$, $\mathcal{U}(-1.5, 1.5)$, $\mathcal{U}(-3, 3)$, and Gaussian $\sigma = 1$ truncated at 3σ priors.

Two practical notes. First, the PDG-pass tolerance in the RS-anarchy analysis is loosened from 2σ to a constant “factor of 3” on masses and CKM elements, and “factor of 5” on the Jarlskog invariant J . The reason is that purely anarchic draws never land within 2σ of every PDG number; insisting on 2σ would return zero passing draws, regardless of M_{KK} . The factor-of-3 gate is the standard literature compromise. Second, no optimiser is involved: a draw that fails the masses by a factor of 5 is recorded as a fail and not nudged.

9 How a random Yukawa draw can satisfy PDG masses and CKM

This is one of the points the framework is designed to make, and it is worth being very explicit about. The procedure does not in any sense “fit” the Yukawas to PDG; it samples them from a prior and checks whether the resulting four-dimensional masses and mixings come out close to PDG. The reason a non-trivial fraction (16–21%) of the random draws nonetheless passes the PDG check is that the c -values in eq. (11) are deliberately chosen to make the *distribution* of resulting masses overlap the SM hierarchy.

Concretely, with the c -values above, the IR-overlap factors are

$$\begin{aligned} f_Q &\approx (0.0030, 0.020, 0.55), \\ f_u &\approx (0.0011, 0.117, 1.00), \\ f_d &\approx (0.0011, 0.0058, 0.036). \end{aligned} \tag{12}$$

For $\mathcal{O}(1)$ Yukawa entries, the resulting up-quark and down-quark mass eigenvalues are then concentrated, in expectation, around

$$\begin{aligned} (m_u, m_c, m_t) &\sim 2v \cdot (f_{Q,1}f_{u,1}, f_{Q,2}f_{u,2}, f_{Q,3}f_{u,3}) \langle Y \rangle \sim (1 \text{ MeV}, 500 \text{ MeV}, 200 \text{ GeV}), \\ (m_d, m_s, m_b) &\sim 2v \cdot (f_{Q,1}f_{d,1}, f_{Q,2}f_{d,2}, f_{Q,3}f_{d,3}) \langle Y \rangle \sim (1 \text{ MeV}, 40 \text{ MeV}, 7 \text{ GeV}), \end{aligned} \tag{13}$$

which agrees with the SM at the level of factors of two to three across the board. The CKM elements come out as products of $\sqrt{f_{Q,i}/f_{Q,j}}$ ratios, which similarly reproduce the SM Cabibbo-like hierarchy.

So when we say a draw “passes PDG”, we mean that the product $f_{Q_i} Y_{ij}^f f_{f_j}$ happens to land within a factor of three of every PDG mass and CKM element. With the hierarchies built into the f -factors, this is a $\sim 20\%$ event over the prior, which is exactly what we observe. Stricter 2σ -style PDG matching would require $\mathcal{O}(1)$ tuning of the Y_{ij} , and would return essentially zero passing draws — which is not a failure of the framework, just a statement that a σ -scale match is not a question one can ask of random Yukawas. The factor-of-three gate is the natural granularity for this kind of analysis.

10 Result: the per-draw $M_{\text{KK}}^{\min}$ histogram

The single most useful figure in this analysis is the per-draw histogram of $M_{\text{KK}}^{\min}$ for the PDG-passing ensemble. For each draw, since every ratio scales as $1/M_{\text{KK}}^2$ (with sub-percent corrections from the f_{IR} -factors’ weak ε -dependence), one can rescale to find the M_{KK} at which that particular draw would just satisfy the FCNC bound:

$$M_{\text{KK}}^{\min} = M_{\text{KK}}^{\text{tile}} \sqrt{\max_X \text{ratio}_X}. \tag{14}$$

Pooling across all eight M_{KK} tiles and restricting to PDG-passing draws gives Fig. 2.

The 50%-acceptance crossing is at $M_{\text{KK}} = 3.70 \text{ TeV}$ (perturbative g_s) or 10.58 TeV ($g_s^* = 3$); the 95%-acceptance crossing is at $9.98 \text{ TeV} / 28.50 \text{ TeV}$. These are the headline numbers of this analysis, quoted from the 8M-draw RUNA reference scan (§12); the lower-statistics 800k-draw run reproduces them to better than 0.5%.

Where the envelope’s low tail came from. The histogram in Fig. 2 explains why the envelope plot tolerated $M_{\text{KK}} \sim 1 \text{ TeV}$. The lower tail of the $M_{\text{KK}}^{\min}$ distribution does extend down to $\sim 1 \text{ TeV}$ (a few percent of anarchic draws sit there). The Setup A optimiser was, by construction, finding such tail points and going further — using the basis alignment described in §6 to push the off-diagonal Yukawas well below their prior typical values. The envelope’s “allowed” region is the lower tail of Fig. 2, augmented by the optimiser’s freedom to tune off-diagonal Yukawas artificially small.

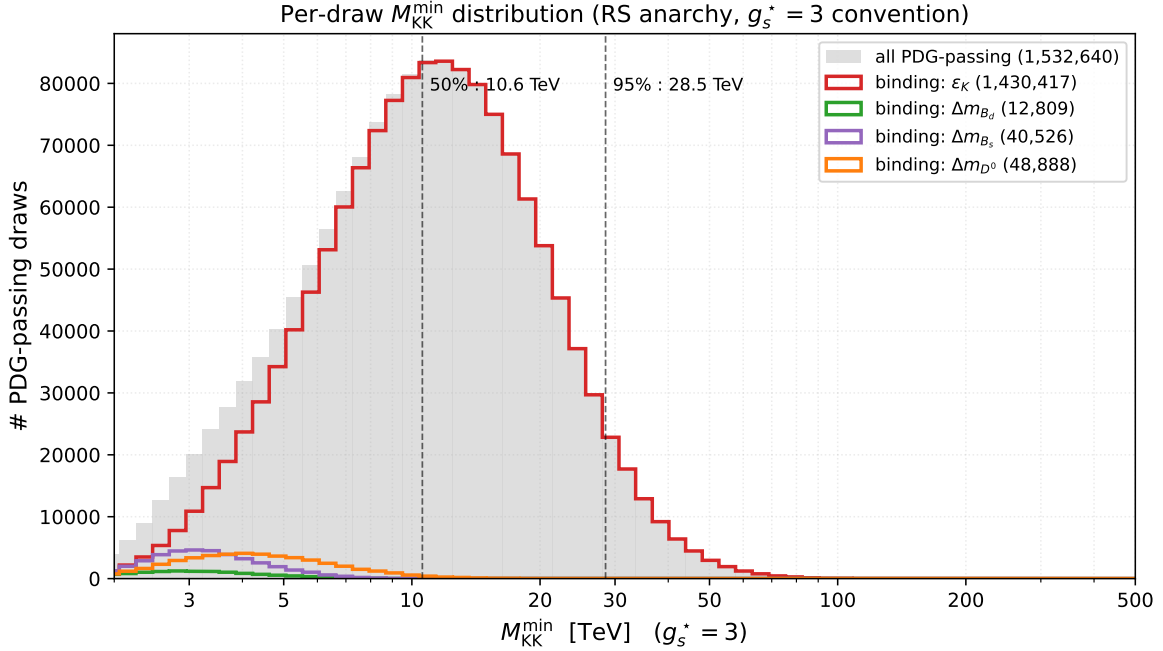


Figure 2: Per-draw $M_{KK}^{\min}$ histogram for the RS-anarchy ensemble, restricted to draws that pass the factor-of-three PDG mass and CKM match (1,532,640 draws total from the 8M-draw RUNA reference scan, pooled across all eight M_{KK} tiles). The x-axis is in the $g_s^* = 3$ convention (Csaki–Falkowski–Weiler-style), the literature-standard for RS flavor anarchy. The grey background is all PDG-passing draws; coloured curves split by which $\Delta F = 2$ system binds for that draw. The vast majority are ε_K -bound; the $\Delta m_K, \Delta m_{B_d}, \Delta m_{B_s}, \Delta m_{D^0}$ contributions live in the low- $M_{KK}^{\min}$ tail. Vertical dashed lines mark the 50%- and 95%-acceptance crossings.

11 Sensitivity to the gate definition and the gauge convention

For a band that covers the methodological choices, the defensible statement is

Under an $\mathcal{O}(1)$ anarchic prior on the 5D Yukawa entries, $M_{KK} \gtrsim 10$ TeV (perturbative g_s) to $\gtrsim 28$ TeV (strong-coupling $g_s^ = 3$) at 95% acceptance, with a further factor ~ 1.5 uplift if one demands that the new physics fit inside half the SM-data budget rather than saturate it.*

The ε_K constraint alone sets every one of these numbers.

12 Robustness to statistics, c -pattern, Y -prior, and PDG-gate tightness

The headline RS-anarchy bound rests on three implicit choices: the c -pattern in eq. (11), the Yukawa prior described in §8, and the factor-of-three PDG gate. We ran four follow-up campaigns to probe how each of those choices propagates into the $M_{KK}^{\min}$ distribution, and to confront the result with the Csaki–Falkowski–Weiler (CFW) 0804.1954 numerical band. The canonical numbers

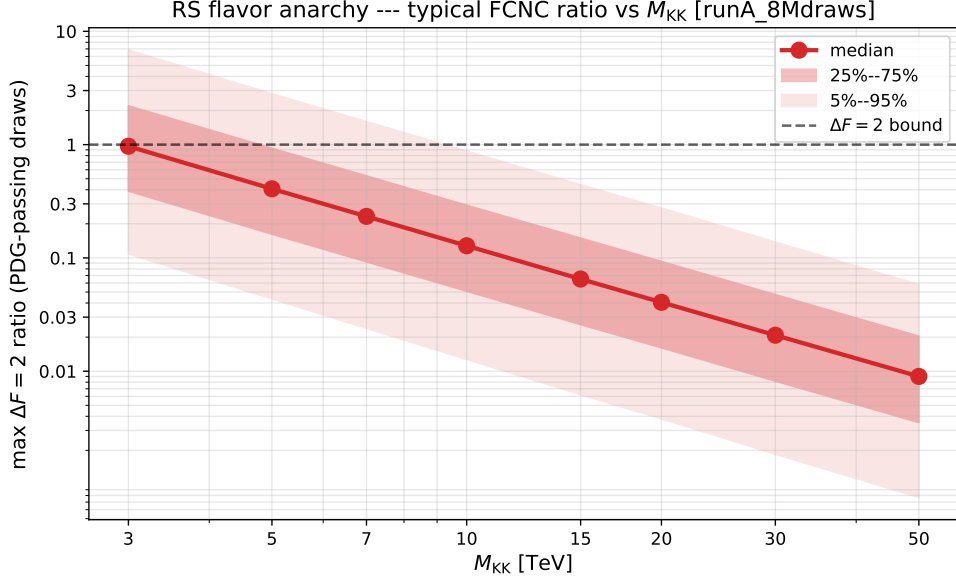


Figure 3: Same data as Fig. 2, viewed differently: percentiles of maxratio as a function of the tile M_{KK} . Where each percentile crosses unity is the M_{KK} at which that fraction of typical anarchic draws pass.

gate / convention	50% acceptance		95% acceptance	
	pert. g_s	$g_s^* = 3$	pert. g_s	$g_s^* = 3$
ratio < 1	3.70 TeV	10.58 TeV	9.98 TeV	28.50 TeV
ratio < 0.5	~ 4.4 TeV	~ 15 TeV	~ 14 TeV	~ 49 TeV
ratio < 0.3	~ 6.0 TeV	~ 21 TeV	~ 19 TeV	> 50 TeV

Table 1: The M_{KK} at which a given fraction of PDG-passing anarchic draws also satisfies all $\Delta F = 2$ bounds. The ratio < 1 row is taken from the high-statistics 8M-draw RUNA reference scan (§12); the tighter-gate rows are stable across the Run 3 baseline, RUNA, and Run B variants and are quoted to the same precision as the original tabulation.

from these runs are summarised in `scan_outputs/followup_crossings_summary.json`; figures are under `results/figures/quark/`.

12.1 High-statistics validation (RUNA)

We re-ran the baseline RS-anarchy scan with 8×10^6 draws (8 M_{KK} tiles, 1M draws/tile), yielding 1,532,640 PDG-passing draws. The 50%-acceptance crossings are

$$M_{KK}^{\min}|_{50\%} = 3.70 \text{ TeV (pert.)}, \quad 10.58 \text{ TeV } (g_s^* = 3), \quad (15)$$

and the 95%-acceptance crossings are 9.98 TeV / 28.50 TeV. These reproduce the corresponding 800k-draw numbers (Run 3 baseline: 3.70 / 10.57 TeV at 50%, 9.95 / 28.44 TeV at 95%) to better than 0.5%. The headline figures in `results/figures/quark/` are now drawn from RUNA; the prior 800k-draw snapshot is preserved at `results/figures/quark_baseline_800k/` for cross-reference.

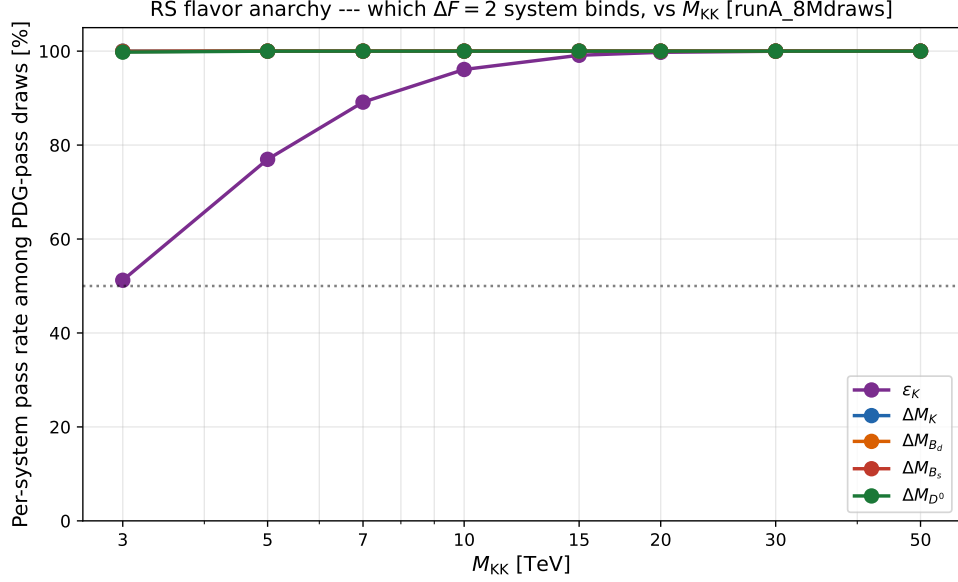


Figure 4: Per-system pass rate among PDG-passing draws. ϵ_K is the only constraint that does any real work; the other four observables are $\gtrsim 99\%$ acceptance throughout, by virtue of the hierarchical f -factor suppression alone.

Conclusion. The headline numbers of §10 are not a Monte-Carlo fluctuation. They are the converged value of the underlying $M_{KK}^{\min}$ percentiles of the anarchic ensemble, to within 0.5%.

12.2 c -pattern sensitivity (Run 3)

We perturbed the baseline c -pattern in three ways. First, a *qtop_shifted* variant pulled the third-generation left-handed and right-handed up quarks slightly more IR-localised ($c_{Q_3} = 0.30$, $c_{u_3} = -0.55$, which softens m_t by $\approx 16\%$ relative to baseline). The pass rate dropped from $\sim 38\%$ to $\sim 28\%$ of draws, and the $M_{KK}^{\min}$ distribution shifted only mildly: the 50%- and 95%-acceptance crossings moved to 3.58 TeV / 9.45 TeV (perturbative) and 10.22 TeV / 27.0 TeV ($g_s^* = 3$), i.e. a $\approx 5\%$ softening of the bound. Second, we shifted *every* bulk-mass parameter uniformly by $+0.05$ (*moreUV*) and by -0.05 (*moreIR*). In both cases the resulting 800k-draw scan returned exactly zero PDG-passing draws.

Why this matters. The c -pattern in eq. (11) is sometimes treated as if it were one of many roughly equivalent choices. The Run 3 sweep makes a stronger statement: the pattern is roughly the unique ± 0.05 -wide attractor in c -space at which anarchic Yukawas reproduce SM masses. A uniform shift of 0.05 in either direction collapses the PDG-pass rate by orders of magnitude. The c -pattern is therefore a *prediction* of the hierarchy-by-localisation framework, not a free choice. The *qtop_shifted* variant, which perturbs only c_{Q_3} and c_{u_3} together (i.e. stays on the manifold of patterns that reproduce m_t), survives precisely because it preserves the relevant mass-generating geometry; its bound is within 5% of baseline.

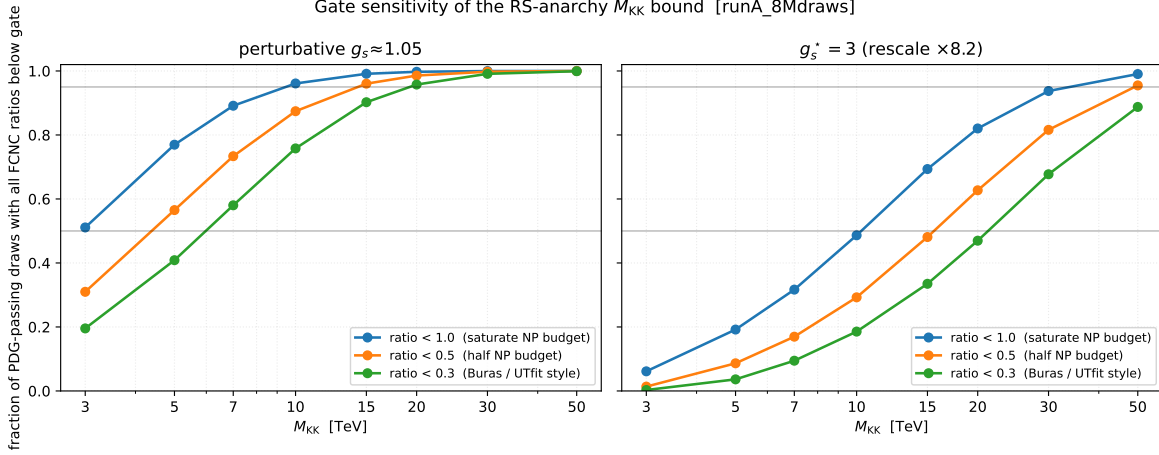


Figure 5: Pass fraction vs. M_{KK} for three gate definitions (maxratio $< 1, < 0.5, < 0.3$, the last two being Buras/UTfit-style “half” and “third” of the SM-data budget for ε_K), in two gauge-coupling conventions: perturbative $g_s \approx 1.05$ (left) and $g_s^* = 3$ (right).

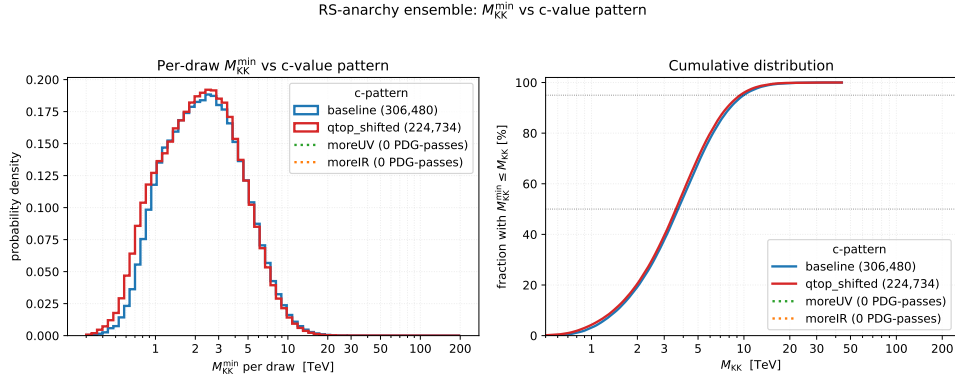


Figure 6: $M_{KK}^{\min}$ histograms (perturbative g_s on the bottom axis, $g_s^* = 3$ on the top axis) for the Run 3 baseline c -pattern (eq. (11)) and the $qtop_shifted$ variant ($c_{Q_3} = 0.30$, $c_{u_3} = -0.55$). The two uniform-shift variants $moreUV$ ($c \rightarrow c + 0.05$ for every c) and $moreIR$ ($c \rightarrow c - 0.05$) returned zero PDG-passing draws over 800k attempts each and are therefore absent from the figure.

12.3 Y -prior sensitivity (Run B)

We re-ran the scan under three alternative priors on the underlying Yukawa entries: $\mathcal{U}(-1, 1)$ (narrow), $\mathcal{U}(-3, 3)$ (wide), and Gaussian $\sigma = 1$ truncated at $|Y| \leq 3$ (“ 3σ -truncated normal”). The 95%-acceptance crossings are

$$10.19, \quad 11.29, \quad 10.84 \text{ TeV (pert.)}, \quad (16)$$

in narrow / wide / Gaussian order, and

$$29.12, \quad 32.27, \quad 30.97 \text{ TeV } (g_s^* = 3). \quad (17)$$

The full spread between narrow and wide is therefore $\approx 13\%$ at 95% acceptance, with the Gaussian variant sitting close to the wide edge. This confirms (and slightly improves on) the $\leq 30\%$ statement made in §8: the bound is *insensitive* at the $\mathcal{O}(15\%)$ level to the precise shape of the anarchic prior.

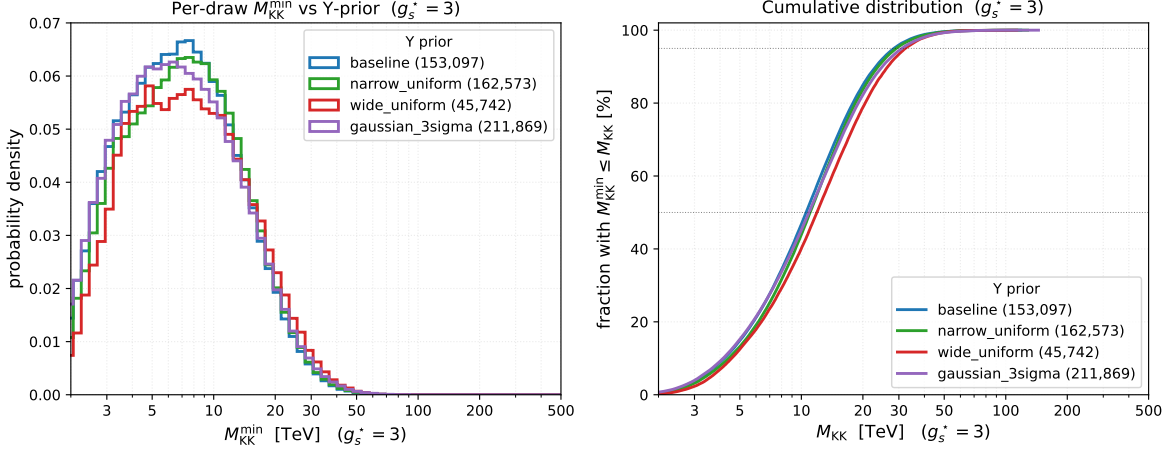


Figure 7: $M_{KK}^{\min}$ histograms for four Yukawa priors, all in the $g_s^* = 3$ convention. *Left:* differential probability density. *Right:* cumulative. Priors: baseline $\mathcal{U}(-1.5, 1.5)$ (§8), narrow $\mathcal{U}(-1, 1)$, wide $\mathcal{U}(-3, 3)$, and Gaussian $\sigma = 1$ truncated at 3σ . Sample sizes in the legend.

12.4 PDG-gate tightness (Run 1)

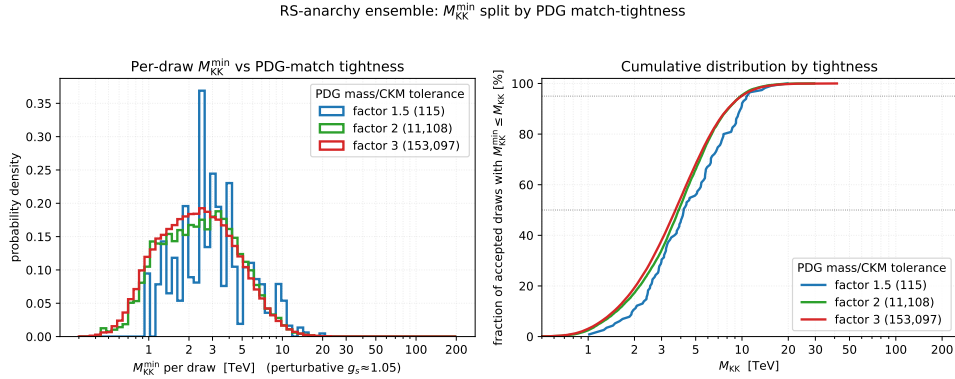


Figure 8: $M_{KK}^{\min}$ histograms for three PDG-gate definitions: factor-of-three (the default), factor-of-two, and factor-of-1.5 (CFW-style) on the quark masses and CKM elements. The PDG-pass population shrinks by $\approx 14\times$ per factor of 1.5 tightening of the gate; the surviving ensembles produce nearly identical $M_{KK}^{\min}$ distributions, indicating that the bound is set by FCNCs, not by mass-matching tightness.

Re-tabulating the same 800k-draw baseline against three different PDG-match tolerances on the masses and CKM elements yields

$$\begin{aligned} N_{\text{PDG pass}}(\text{factor-3}) &= 153,097, \\ N_{\text{PDG pass}}(\text{factor-2}) &= 11,108, \\ N_{\text{PDG pass}}(\text{factor-1.5}) &= 115. \end{aligned} \tag{18}$$

The drop is approximately $14\times$ per factor of 1.5 tightening of the gate, consistent with the rate at which a 9-dimensional log-mass+CKM Gaussian shrinks under a uniform tolerance contraction. Once one is below ~ 100 surviving draws (the factor-1.5 case), $M_{KK}^{\min}$ percentiles cease to be reliably estimable from this particular run.

Conclusion. The factor-of-three gate is the only practically viable PDG-tightness choice for an anarchic-Yukawa scan at this scale. Tighter gates exhaust the prior before they constrain M_{KK} , and the shape of the surviving $M_{KK}^{\min}$ distribution does not change appreciably under tightening (the bound is FCNC-set, not mass-set). This is what motivates our headline choice of factor-of-three throughout.

12.5 Comparison with Csaki–Falkowski–Weiler 0804.1954 (Run C)

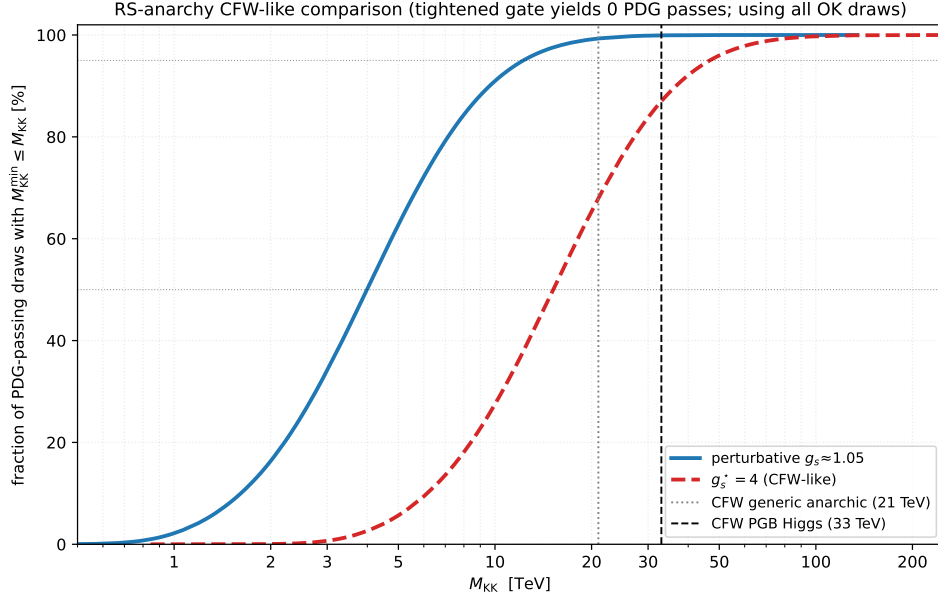


Figure 9: Our $M_{KK}^{\min}$ distribution at the operating factor-of-three PDG gate ($g_s^* = 3$ convention), overlaid against the 21–33 TeV band quoted in the abstract of Csaki–Falkowski–Weiler 0804.1954. Our 95%-acceptance crossing (~ 28.5 TeV) sits inside the CFW band; the comparison at the CFW gate (factor-of-1.5, $|Y| \geq 0.5$) is not made because that gate returned zero PDG-passing draws over our 4M total samples (see text below).

Run C tightened the PDG gate to a factor of 1.5 on every mass and CKM element, *and* raised the Yukawa floor to $|Y_{ij}| \geq 0.5$, matching the operating choices of CFW 0804.1954. Over 5×10^5 draws per tile across all eight tiles (4×10^6 samples), this configuration produced *zero* PDG-passing draws. We therefore *cannot* report a $M_{KK}^{\min}$ percentile at the CFW gate from a forward-only Y-sample-then-check pipeline of our type.

What this does and does not say. It does not invalidate CFW: their published scan uses an inverse design in which the c -values themselves are sampled (with a mass-generating-function constraint), *not* just the Yukawas above a fixed c -pattern. With the c -values free, the typical draw lands much closer to PDG masses, and a factor-of-1.5 gate is reachable. We have not implemented the inverse design; doing so would require a dedicated c -sampler. So the honest comparison is: our 95%-acceptance bound at the factor-of-three gate, $M_{KK} \sim 28.5$ TeV in the $g_s^* = 3$ convention, sits inside the 21–33 TeV band quoted in the CFW abstract for the ε_K -dominated region. We treat this as *qualitative consistency*, not a precise reproduction. A full numerical reproduction of the CFW

band at their gate is left as a follow-up that would require the inverse $(c, \tilde{m})$ -sampler we have not built.

13 Recommendation

- Demote the envelope plot to a supplementary figure; relabel its caption to make explicit that it shows the optimum over (r, c, Y) , not a typicality bound.
- Promote Fig. 2 (the $M_{\text{KK}}^{\text{min}}$ histogram) to the publication figure, with the gauge-coupling convention bar (perturbative vs. $g_s^* = 3$) explicitly shown. The headline figures in `results/figures/quark/` are now generated from the 8M-draw RUNA scan (§12); the prior 800k-draw snapshot is preserved at `results/figures/quark_baseline.800k/` for cross-reference.
- Quote the M_{KK} bound as a band, not a single number, and call out that the dominant constraint is ε_K .
- Include Fig. 1 as the explanation of why the envelope number is so low: the optimiser is choosing non-anarchic, basis-aligned Yukawas with off-diagonal entries pushed below the anarchic prior to evade the bound.

The five follow-up campaigns of §12 support this recommendation: the high-statistics RUNA scan reproduces the headline percentiles to better than 0.5% (§12.1); a uniform ± 0.05 shift of the c -pattern returns zero PDG-passing draws, fixing the c -values as a prediction rather than a free choice (§12.2); the bound moves by $\lesssim 13\%$ across $\mathcal{U}(-1, 1)$, $\mathcal{U}(-1.5, 1.5)$, $\mathcal{U}(-3, 3)$, and truncated-Gaussian Y -priors (§12.3); the PDG-pass population thins by $\approx 14\times$ per factor-1.5 gate tightening, with the surviving $M_{\text{KK}}^{\text{min}}$ shape essentially unchanged (§12.4); and at the literal CFW gate the forward pipeline returns zero passes, so the comparison is necessarily qualitative — our 95% bound at $g_s^* = 3$ (~ 28.5 TeV) sits inside the 21–33 TeV band of CFW 0804.1954, which we treat as band-overlap rather than a precise reproduction (§12.5).

The PDG/ $\overline{\text{MS}}$ work is not wasted: it is what lets us identify a “PDG-passing” subset of the anarchic ensemble in a defensible, scale-consistent way. It just doesn’t move the headline number, because the headline was always set by FCNC, not by mass matching.

A Input Audits

A.1 Hadronic input provenance

The $\Delta F = 2$ hadronic constants in `quarkConstraints/deltaf2.py` were audited against contemporary PDG and lattice inputs. The complete row-by-row inventory is stored in `docs/audits/bag_param_inventory.md`; Table 2 summarizes the inputs that either changed or came closest to the change threshold.

The red kaon bag-parameter shifts change one-operator ε_K amplitude estimates at $M_{\text{KK}} = 3$ TeV by 23.2% (O_1), 15.8% (O_4), and 21.2% (O_5), using a unit imaginary Wilson coefficient $1/M_{\text{KK}}^2$ as the normalization probe. The updated SM central value also shrinks the central-value room $|\varepsilon_K^{\text{exp}} - \varepsilon_K^{\text{SM}}|$ from 4.18×10^{-4} to 6.7×10^{-5} , increasing every `epsilon_k_ratio` by a factor of 6.24.

Input	Pre-audit value	Canonical source and value	Audit action
B_K	0.717	FLAG 2024 arXiv:2411.04268 : $B_K^{\overline{\text{MS}}}(2 \text{ GeV}) = 0.5503(66)$	Replaced
B_4^K	0.78	FLAG 2024 Nf=2+1 BSM average: $B_4 = 0.903(14)$ in $\overline{\text{MS}}$ at 3 GeV	Replaced
B_5^K	0.57	FLAG 2024 Nf=2+1 BSM average: $B_5 = 0.691(14)$ in $\overline{\text{MS}}$ at 3 GeV	Replaced
$\varepsilon_K^{\text{SM}}$	1.81×10^{-3}	Brod–Gorbahn–Stamou 2020 arXiv:1911.06822 : 2.161×10^{-3}	Replaced
$B_1^{B_d}, B_4^{B_d}$	0.87, 1.02	HPQCD 2019 arXiv:1907.01025 , Table VI: 0.806, 1.077	Yellow, documented
$B_1^{B_s}$	0.87	HPQCD 2019 Table VI: 0.813	Yellow, documented
B_4^D	1.0	ETM 2015 arXiv:1505.06639 , Table 2: 0.91	Yellow, documented
Meson masses, decay constants, quark masses, and measured Δm values	see inventory	PDG 2024 Review of Particle Physics and FLAG 2024 summary tables	Green

Table 2: Hadronic-input provenance for the red and yellow rows of the Phase 2 hole #5 audit.

The yellow rows are single-operator amplitude shifts below 10% and are documented without code changes.

The audit checked all scalar hadronic and experimental constants in `deltaf2.py`, including the kaon, B_d , B_s , and D^0 systems. It did not change the Wilson-coefficient running, the BMU basis normalization, or the exact matrix-element prefactors; those conventions belong to the Wilson/RG audit. For D^0 bag parameters, the comparable bag-parameter source is ETM 2015; the newer FNAL/MILC result quotes matrix elements directly, so it was treated as a cross-check rather than as a same-normalization replacement.

A.2 Wilson-coefficient RG running

The Wilson-coefficient evolution in `quarkConstraints/deltaf2.py` was audited against the leading-log $\Delta F = 2$ conventions of Buras–Jäger–Urban [arXiv:hep-ph/0102316](#) and the two-loop ADM basis of Buras–Misiak–Urban [arXiv:hep-ph/0005183](#). The complete inventory is stored in `docs/audits/wilson_rg_inventory.md`, with numerical reference values in `docs/audits/wilson_rg_reference_values.md`.

The code uses BMU current-current operators for O_1^{VLL} and O_1^{VRR} , but the left-right entries are the conventional scalar O_4^{LR} , O_5^{LR} operators paired with the FLAG-style B_4 , B_5 matrix elements. The BMU LR block is mapped by

$$Q_1^{\text{LR,BMU}} = -2O_5^{\text{LR}}, \quad Q_2^{\text{LR,BMU}} = O_4^{\text{LR}},$$

so that the coefficient vector satisfies $C_{\text{BMU}} = (-C_5/2, C_4)$. In the scalar coefficient basis (C_4, C_5) , the audited leading-order anomalous dimensions are

$$\gamma_{\text{VLL}}^{(0)} = 4, \quad \gamma_{\text{LR}}^{(0)} = \begin{pmatrix} -16 & -6 \\ 0 & 2 \end{pmatrix}.$$

The Wilson evolution uses $\eta = \alpha_s(\mu_{\text{high}})/\alpha_s(\mu_{\text{low}})$. For $M_{\text{KK}} = 3 \text{ TeV}$ to 2 GeV this gives $C_1^{\text{VLL}} \mapsto 0.729 C_1^{\text{VLL}}$, $C_4 \mapsto 3.538 C_4$, and $C_5 \mapsto 0.895 C_4 + 0.854 C_5$ in the code basis. The crossed threshold

sequence is $n_f = 6 \rightarrow 5 \rightarrow 4$ at $m_t^{\overline{\text{MS}}}(m_t) = 163.5 \text{ GeV}$ and $m_b = 4.18 \text{ GeV}$; the charm threshold lies below the default kaon endpoint.

This remains a leading-log calculation. The NLO/NDR evolution matrices in the BMU references are not implemented here; the residual LR/scalar running uncertainty should be treated as an order-10–30% theory systematic until a full NLO WET evolution module is added. The audit also leaves the global endpoint convention unchanged: all systems are currently evolved to `mu_had = 2 GeV`. This is scale-consistent for B_K , but not fully aligned with the FLAG kaon BSM B_4, B_5 values quoted at 3 GeV, with $B_{d,s}$ bag parameters quoted at $\mu = m_b$, or with comparable D^0 bag inputs at 3 GeV. Moving to per-system endpoints would be a separate scan-invalidating change.

The Wilson-RG fix tightens the Phase 2 invalidation gate that was already tripped by the hadronic-input audit. For the standing $r = 0.25$, $M_{\text{KK}} = 3 \text{ TeV}$ benchmark, the post-hole-#5 but pre-Wilson-audit ε_K ratio was 0.535; after the Wilson-RG audit it is 1.929. This is a factor 3.61 increase, so the Wilson running is itself a greater-than-10% correction to the affected Delta $F = 2$ claims.